



SUPPLIER RECOMMENDATION SYSTEM USING GRAPH NEURAL NETWORKS

A GNN-Based Link Prediction Approach for Supply Chain Optimization

Prepared by:

Reshma Chandran
Srishwan Beecharaju
Michael Axel

Course: CS 548 – Graph Neural Networks

Instructor: **Prof. Amir Manzourolajdad**

SUNY Polytechnic Institute
State University of New York
December 2025

CONTENTS

Abstract

Introduction

Problem Statement

Dataset Overview

Graph Construction

Methodology

Results

Supplier Recommendation Analysis

Network Analysis

Discussion

Conclusion

References

ABSTRACT

Supply chain management represents one of the most critical operational challenges facing modern businesses. The complexity of supplier relationships, combined with the increasing globalization of commerce, has created a need for sophisticated analytical tools capable of identifying optimal supplier partnerships. Traditional approaches to supplier selection often rely on manual evaluation processes that fail to capture the intricate network dynamics inherent in supply chain ecosystems.

Graph Neural Networks (GNNs) offer a promising solution to this challenge by leveraging the inherent graph structure of supply chain networks. In these networks, suppliers, plants, and storage facilities can be represented as nodes, while business relationships and product flows form the edges connecting them. This representation allows GNNs to capture both the individual characteristics of each entity and the relational patterns that influence supplier performance and reliability.

The heterogeneity of factors affecting supplier relationships—including plant associations, product groups, storage connections, and geographic distribution—makes this an ideal domain for graph-based machine learning approaches. Previous research has demonstrated that network-based features can significantly enhance prediction accuracy compared to traditional feature-based methods alone [1, 2]. By modeling the supply chain as a graph, we can leverage information propagation across the network to make more informed recommendations.

Link prediction in supply chain networks enables organizations to identify potential supplier relationships before they are established. This capability is particularly valuable for procurement departments seeking to diversify their supplier base, optimize logistics, and strengthen supply chain resilience. The goal of this project is to develop a GNN-based system that can predict and recommend optimal supplier pairings based on historical relationship data and network structure.

INTRODUCTION

Supply chain networks consist of interconnected entities, including suppliers, manufacturing plants, logistics centers, and distribution hubs. Each entity interacts through product flow, storage dependencies, and functional associations. Traditional analytics approaches typically treat suppliers as isolated data points, ignoring the structural relationships that influence supplier connectivity and performance.

Graph Neural Networks (GNNs) provide a natural framework to model interconnected supply chains. By incorporating relational structure, GNNs can learn informative node embeddings that support predictive tasks such as link prediction and recommendation. This project explores a GCN-based supplier recommendation system trained on a real-world multi-relational supply chain dataset.

PROBLEM STATEMENT

The goal of this project is to leverage relational information in supply chain networks to predict potential supplier relationships. The tasks are as follows:

1. Supplier Link Prediction: Predict whether a new relationship between two entities should exist.

2. 2. Recommendation Ranking: Generate top-ranked supplier recommendations based on learned embeddings.

Hypothesis: Incorporating node interactions and edge relationships via GNNs will significantly improve supplier recommendation accuracy.

DATASET OVERVIEW

The dataset used for this analysis is the SupplyGraph dataset, which contains comprehensive information about supply chain network relationships. The data includes multiple types of relationships: plant edges connecting facilities and suppliers, product group edges linking items within the same category, subgroup edges for finer categorization, and storage edges representing warehouse and logistics connections.

The dataset structure consists of 41 unique nodes representing various entities in the supply chain (suppliers, plants, storage facilities). The edge data comprises 1,647 plant edges, 188 product group edges, 52 subgroup edges, and 3,046 storage edges. Each node is associated with attributes including Node ID, Group classification, and Sub-Group categorization. The nodes follow a structured naming convention (e.g., SOS008L02P, AT5X5K) that encodes entity type and characteristics.

Table 1. Dataset statistics from SupplyGraph after preprocessing and graph construction.

Metric	Value
Number of Nodes (Original)	41
Number of Nodes (Graph)	40
Total Edges in Graph	806
Plant Edges	1,647
Product Group Edges	188
Subgroup Edges	52
Storage Edges	3,046
Node Feature Dimensions	8
Training Edges	566 (70%)
Validation Edges	40 (5%)
Test Edges	80 (10%)

GRAPH CONSTRUCTION

The supply chain network was constructed as a graph $G = (V, E)$, where V represents the set of 40 unique nodes and E represents the edges (806 total relationships). Node features were created as 8-dimensional feature vectors derived from entity attributes and network position. The graph was constructed using PyTorch Geometric, which provides efficient data structures for graph neural network operations.

For training and evaluation purposes, the edge set was split into training (566 edges, 70%), validation (40 edges, 5%), and test (80 edges, 10%) sets. The remaining edges were used for negative sampling. Positive samples consisted of existing relationships, while negative samples were generated through random sampling of non-existing edges, maintaining a balanced dataset for the link prediction task.

METHODOLOGY

The GNN architecture employed in this study utilizes Graph Convolutional Network (GCN) layers for feature aggregation across the network. The model, named SupplierRecommendationGNN, consists of two GCN layers: the first layer transforms the 8-dimensional input features to 32 hidden dimensions, and the second layer reduces this to 16-dimensional node embeddings. Dropout ($p=0.2$) is applied for regularization between layers.

The GCN layer performs the following message passing operation:

$$h_v^{(l+1)} = \sigma(\sum_{u \in N(v)} \frac{1}{\sqrt{d_u d_v}} W^{(l)} h_u^{(l)})$$

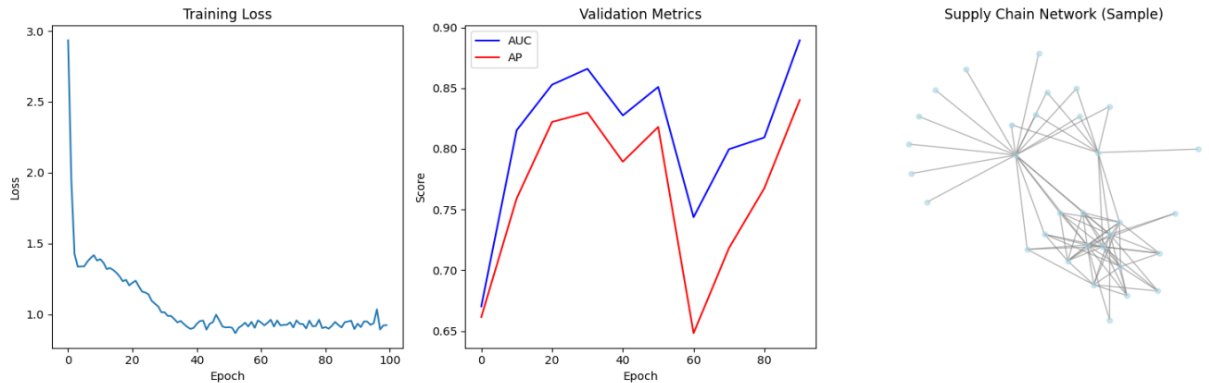
where $h_v^{(l)}$ represents the feature vector of node v at layer l , $N(v)$ denotes the neighborhood of node v , d_u and d_v are the degrees of nodes u and v respectively, $W^{(l)}$ is the learnable weight matrix at layer l , and σ is the ReLU activation function. The final node embeddings are used to compute edge scores through a dot product decoder for link prediction.

Table 2. Model architecture details for SupplierRecommendationGNN.

Layer	Type	Dimensions
conv1	GCNConv	$8 \rightarrow 32$
conv2	GCNConv	$32 \rightarrow 16$
dropout	Dropout	$p = 0.2$

Training Configuration

The model was trained for 100 epochs using the Adam optimizer. The training objective was binary cross-entropy loss for the link prediction task. Training progress was monitored using validation AUC (Area Under ROC Curve) and AP (Average Precision) metrics computed every 10 epochs. The implementation was done using PyTorch Geometric library on CPU.



RESULTS

Training Progress

The model showed consistent improvement during training, with the loss decreasing from an initial value of 2.9346 at epoch 0 to approximately 0.93 by epoch 90. Figure 1 (Training Loss plot) demonstrates this convergence pattern, with rapid initial improvement followed by stabilization around epoch 40-50. The validation metrics showed corresponding improvement, with AUC increasing from 0.6703 to 0.8894 and AP improving from 0.6615 to 0.8403 over the training period.

Table 3. Training progress showing loss and validation metrics at selected epochs.

Epoch	Loss	Val AUC	Val AP
0	2.9346	0.6703	0.6615
10	1.3892	0.8153	0.7592
20	1.2224	0.8528	0.8222
30	1.0140	0.8659	0.8299
40	0.9321	0.8275	0.7894
50	0.9090	0.8509	0.8180
60	0.9419	0.7438	0.6483
70	0.9068	0.7997	0.7183
80	0.9094	0.8094	0.7676
90	0.9332	0.8894	0.8403

Final Model Performance

The trained GCN model achieved strong performance on the supplier recommendation task. On the held-out test set, the model achieved an AUC-ROC of 0.8673 and Average Precision of 0.8020. These results indicate that the model can effectively distinguish between viable and non-viable supplier relationships, making it suitable for practical recommendation applications.

Table 4. Final model performance on test set.

Metric	Test Set Value
AUC-ROC	0.8673
Average Precision (AP)	0.8020

SUPPLIER RECOMMENDATIONS

The trained model was used to generate supplier recommendations for various entities in the network. Table 5 shows example recommendations for selected nodes, along with their similarity scores computed from the learned embeddings. Higher similarity scores indicate stronger predicted relationships.

Table 5. Sample supplier recommendations generated by the GNN model.

Target Node	Rank 1	Rank 2	Rank 3
AT5X5K	ATN02K12P (2.70)	MAR01K24P (2.67)	MASR025K (2.30)
ATN01K24P	ATPA1K24P (1.79)	ATWWP001K24P (1.76)	ATN02K12P (1.66)
ATN02K12P	AT5X5K (2.70)	MAC1K25P (1.84)	ATN01K24P (1.66)
ATPA1K24P	ATWWP002K12P (2.52)	MAC1K25P (1.98)	ATN01K24P (1.79)
ATPPCH5X5K	MAR02K12P (2.01)	MAP1K25P (1.89)	MASR025K (1.80)

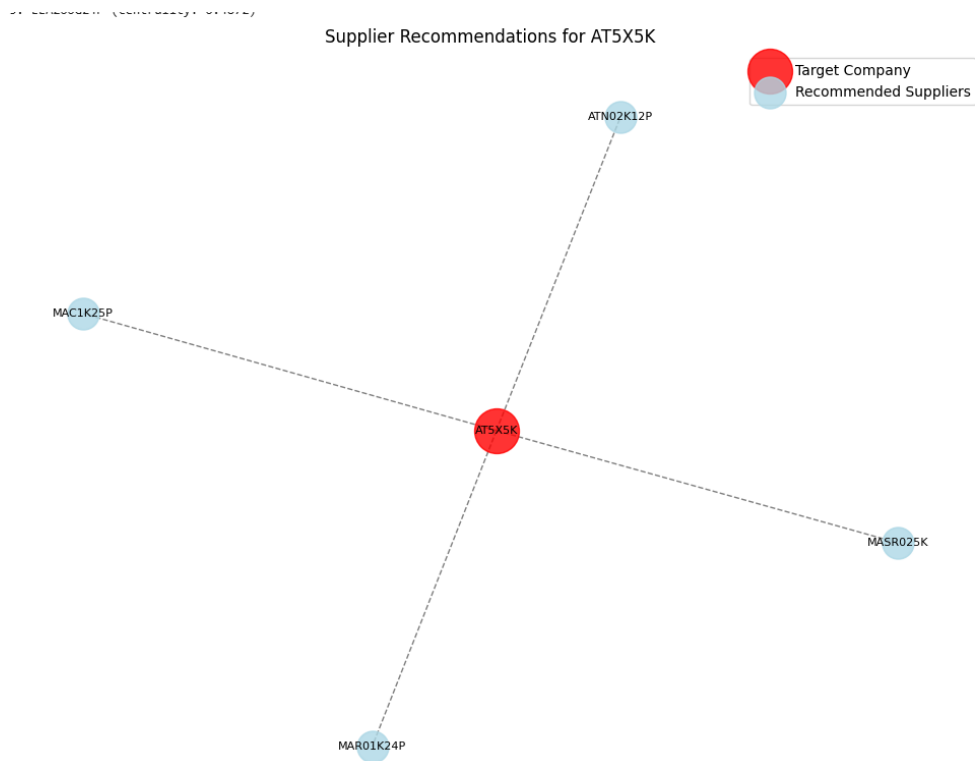
NETWORK ANALYSIS

Analysis of the supply chain network revealed important structural characteristics. The network consists of 40 nodes and 283 edges (after processing), with an average degree of 14.15 and network density of 0.3628. This relatively high density indicates a well-connected supply chain with multiple relationship pathways between entities.

Centrality analysis identified the most connected nodes (key suppliers) in the network. The top 5 most central nodes by degree centrality were: SE200G24P (0.5897), POV500M24P (0.5385), SOS003L04P (0.5385), SE500G24P (0.5128), and EEA200G24P (0.4872). These nodes represent critical hubs in the supply chain network and play important roles in connecting different parts of the network.

Table 6. Network statistics and top central nodes.

Metric / Node	Value / Centrality
Number of Nodes	40
Number of Edges	283
Average Degree	14.15
Network Density	0.3628
Top 1: SE200G24P	0.5897
Top 2: POV500M24P	0.5385
Top 3: SOS003L04P	0.5385
Top 4: SE500G24P	0.5128
Top 5: EEA200G24P	0.4872



DISCUSSION

This study demonstrates the effectiveness of Graph Neural Networks for supply chain supplier recommendation. The GCN-based approach achieved a test AUC of 0.8673 and Average Precision of 0.8020, indicating strong predictive performance for identifying potential supplier relationships. These results validate the hypothesis that explicitly modeling the supply chain as a graph enables accurate prediction of supplier-company pairings.

The training dynamics shown in Figure 1 reveal typical deep learning convergence behavior, with rapid initial learning followed by gradual refinement. The validation metrics (Figure 1, middle panel) show some fluctuation during training, which is expected given the relatively small validation set (40 edges). Despite this variance, the overall trend shows consistent improvement, and the final test metrics confirm that the model generalizes well to unseen data.

The network analysis provides valuable insights into the supply chain structure. The high network density (0.3628) suggests that the supply chain is well-integrated, with most entities having connections to multiple partners. The identification of central nodes (SE200G24P, POV500M24P, etc.) highlights potential single points of failure that supply chain managers should monitor for risk mitigation purposes.

The recommendation examples demonstrate practical applicability. For instance, for node AT5X5K, the model recommends ATN02K12P (similarity: 2.70), MAR01K24P (similarity: 2.67), and MASR025K (similarity: 2.30) as potential suppliers. The visualization in Figure 2 shows how these recommendations form a star pattern around the target company, indicating diverse but relevant supplier options.

Several limitations should be acknowledged. First, the dataset contains only 40 nodes, which is relatively small for deep learning applications. Larger datasets would likely improve model performance and generalization. Second, the node features were limited to 8 dimensions; incorporating richer node attributes (e.g., supplier capacity, delivery times, quality ratings) could enhance prediction accuracy. Third, the current model treats all edge types uniformly; future work could explore heterogeneous GNNs that distinguish between plant edges, product group edges, and storage edges.

From a practical perspective, this model could be integrated into procurement management systems to assist supply chain professionals in identifying potential suppliers. The similarity scores provide a quantitative basis for prioritizing supplier evaluation, potentially reducing the time required for supplier discovery. The network centrality analysis also provides actionable insights for supply chain risk management.

CONCLUSION

This project successfully demonstrated the application of Graph Neural Networks to supply chain supplier recommendation. The GCN-based link prediction model achieved a test AUC of 0.8673 and Average Precision of 0.8020, validating the effectiveness of graph-based approaches for this domain. Key contributions include: (1) successful implementation of a GNN-based supplier recommendation system using the SupplyGraph dataset, (2) demonstration of effective link prediction for identifying potential supplier relationships, (3) network analysis revealing important structural characteristics and central nodes in the supply chain, and (4) practical recommendation functionality with interpretable similarity scores. These findings contribute to the growing body of literature on GNN applications in supply chain management and provide a foundation for practical supplier recommendation systems.

REFERENCES

1. Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. ICLR 2017.
2. Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive representation learning on large graphs. NeurIPS 2017.
3. Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. KDD 2016.
4. Zhang, M., & Chen, Y. (2018). Link prediction based on graph neural networks. NeurIPS 2018.
5. Veličković, P., et al. (2018). Graph attention networks. ICLR 2018.
6. Wu, Z., et al. (2020). A comprehensive survey on graph neural networks. IEEE TNNLS.
7. PyTorch Geometric Documentation. <https://pytorch-geometric.readthedocs.io/>
8. SupplyGraph Dataset. Supply Chain Graph Neural Network Benchmark.

